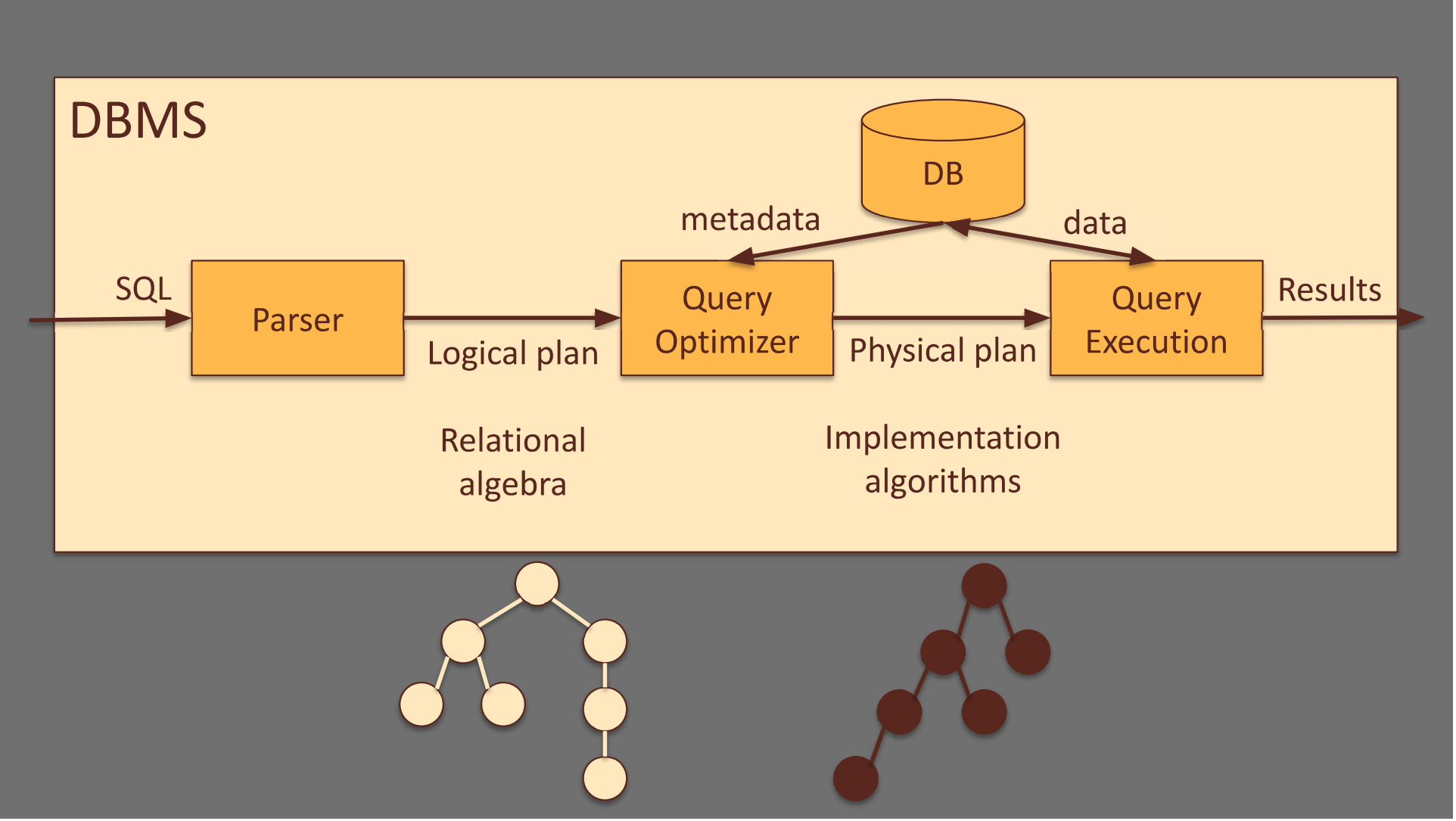
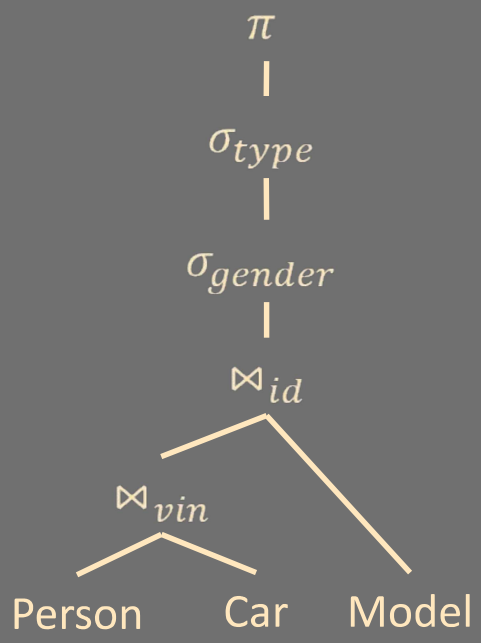
Query Optimization Overview


DBMS
DB
metadata
data
SQL
Parser
Logical plan
Query Optimizer
Physical plan
Query Execution
Results
Relational algebra
Implementation algorithms

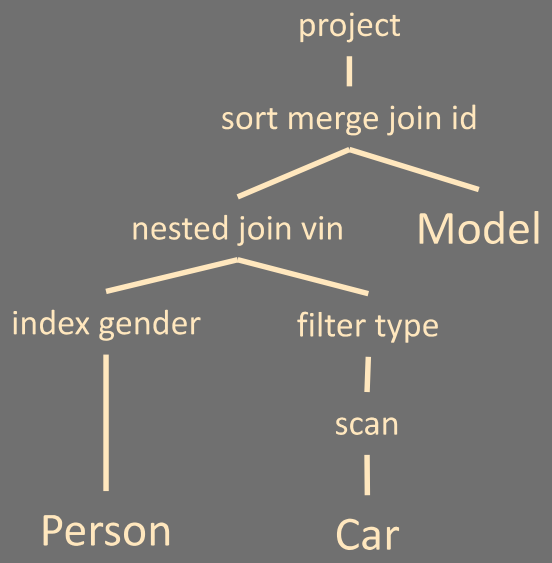
```
SELECT p.first_name,
       p.last_name,
       m.company
FROM Person AS p
INNER JOIN Car AS c ON p.car_vin = c.vin
INNER JOIN Model AS m ON c.model_id = m.id
WHERE p.gender = 'F' AND
      c.type IN ('sedan', 'coupe');
```

Logical plans



```
SELECT p.first_name,
       p.last_name,
       m.company
FROM Person AS p
INNER JOIN Car AS c ON p.car_vin = c.vin
INNER JOIN Model AS m ON c.model_id = m.id
WHERE p.gender = 'F' AND
      c.type IN ('sedan', 'coupe');
```

Physical plans

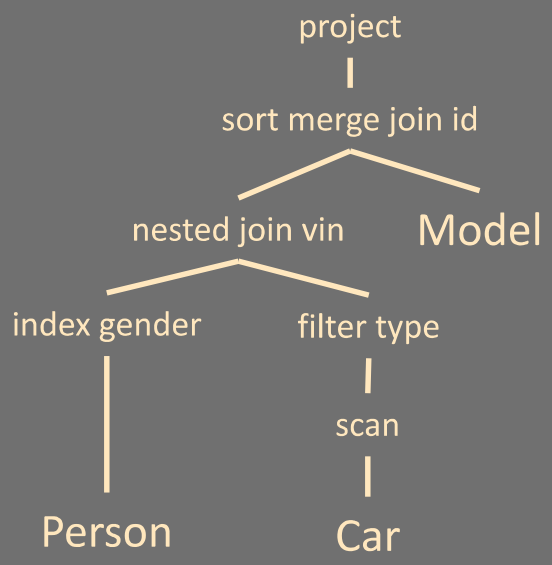


Which plan is fastest?

When running query, know table sizes.

Estimate input sizes.

Estimate operation costs.



Optimization steps

1. Generate all logical plans using relational algebra equivalences
2. For each logical plan, generate all physical plans using possible implementations
3. For each physical plan, estimate costs
4. Pick cheapest physical plan

```
/* TPC_H Query 8 - National Market Share */
SELECT
    O_YEAR,
    SUM(CASE WHEN NATION = 'BRAZIL' THEN VOLUME ELSE 0 END) / SUM(VOLUME)
       AS MKT_SHARE
FROM
    (SELECT
         datepart(yy,O_ORDERDATE) AS O_YEAR,
         L_EXTENDEDPRICE*(1-L_DISCOUNT) AS VOLUME,
         N2.N_NAME AS NATION
     FROM PART, SUPPLIER, LINEITEM, ORDERS, CUSTOMER, NATION N1, NATION N2, REGION
     WHERE P_PARTKEY = L_PARTKEY AND
           S_SUPPKEY = L_SUPPKEY AND
           L_ORDERKEY = O_ORDERKEY AND
           O_CUSTKEY = C_CUSTKEY AND
           C_NATIONKEY = N1.N_NATIONKEY AND
           N1.N_REGIONKEY = R_REGIONKEY AND
           R_NAME = 'AMERICA' AND
           S_NATIONKEY = N2.N_NATIONKEY AND
           O_ORDERDATE BETWEEN '1995-01-01' AND '1996-12-31' AND
           P_TYPE= 'ECONOMY ANODIZED STEEL') AS ALL_NATIONS
GROUP BY O_YEAR
ORDER BY O_YEAR
```

Results in ~22M query plans.
How to quickly find one good enough?

Real-world problems

Too many plans to try them all.

- Heuristics generate only some plans.
- Steps combined so that costs estimates can guide plan generation.

Cost estimates can be wrong.

May generate sub-optimal plan.

Query Optimization

Pat Selinger, IBM System R Project (1970s)

Query optimization is not rocket science. When you flunk out of query optimization, we make you go build rockets.

– Anonymous

What you need to do

- Read a physical query plan.
- Conceptually & experimentally evaluate plan.
- Tune query & DBMS to improve plan.